

Software Requirements Specification (SRS)

Project Title

Command-Line Expense Tracker

1. Purpose

The purpose of this application is to provide users with a simple command-line interface for recording, viewing, searching, summarizing, and deleting personal expenses.

The application is intended to strengthen Python programming skills through practical implementation of file handling, functions, modular programming, error handling, and data processing.

2. Scope

The application will allow users to:

- Add expenses
- View all expenses
- Search expenses by category
- Display monthly summaries
- Delete expenses
- Persist data between executions using a CSV file

The application will operate entirely through the terminal and will not require a graphical interface.

3. Functional Requirements

FR-1 Add Expense

The system shall allow the user to add a new expense.

Required fields:

- Date
- Category
- Description
- Amount

The expense shall be saved to persistent storage.

FR-2 View Expenses

The system shall display all recorded expenses in tabular format.

Displayed columns:

- Date
 - Category
 - Description
 - Amount
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FR-3 Search Expenses

The user shall be able to search expenses by category.

The system shall return all matching expenses.

FR-4 Monthly Summary

The system shall calculate:

- Total spending
- Spending per category

The summary shall be displayed in a readable format.

FR-5 Delete Expense

The user shall be able to delete an expense by selecting its index.

The deleted expense shall no longer appear in storage.

FR-6 Exit Application

The user shall be able to terminate the program gracefully.

4. Non-Functional Requirements

- The application shall start in under one second.
 - Invalid user input shall not crash the application.
 - Code shall be modular and reusable.
 - Data shall persist between executions.
 - The application shall support at least 10,000 expense records.
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5. Assumptions

- One user accesses the application at a time.
 - Data is stored locally.
 - Internet access is not required.
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6. Constraints

- Python standard library only.
 - CSV used as persistent storage.
 - Terminal-based interface only.
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7. Future Enhancements

- Budget tracking
- Expense editing
- Charts and reports
- SQLite database
- User authentication
- Export to Excel
- REST API
- Web interface